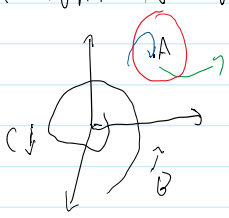
Lecture 3 - 06/05

Wednesday, May 13, 2026 9:03 AM

We saw some examples of dynamical system, we want to analyze them without solving them.

$\dot{X}(t) = F(x(t))$



we have a space
given any point we know where it goes
we can integrate that, but that's hard / impossible, could simulate
or say things about the solutions without solving.
"what happens when $t \to \infty$", goes to ∞? goes to 0? a fixed point? another?

A is a fixed point if $F = 0$ if we begin in that point, it won't move,
finding it is easy, just solve $F(x) = 0$
what happens around it? goes away (unstable)? converges (stable)? goes around?
in general, we can linearize the system near the fixed point, look what happens around it.

linear dynamical systems - $F(t)$ and $X(t)$ are linear

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\omega^2 x - \epsilon x^3 \end{cases}$$

this becomes non linear, this is harder to solve.
in general you can mostly drop/replace the non-linear component near a fixed point

$\underline{X}$ is an n dimentional vector
$\underline{X}(t) = (x_1(t), \dots, x_n(t))$ its linear if all x_i are linear

$$\begin{cases} \dot{x}_1(t) = a_{11} x_1(t) + \dots + a_{1n} x_1(t) \\ \vdots \\ \dot{x}_n(t) = a_{n1} x_1(t) + \dots + a_{nn} x_n(t) \end{cases}$$

we can write a_{ij} as a matrix

$$\boxed{\dot{\underline{X}}(t) = A \cdot \underline{X}(t)}$$

this is an easy case. $X = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is always a fixed point

given 2 solution $\underline{X}^1(t), \underline{X}^2(t) \Rightarrow \underline{X}(t) = c_1 \underline{X}^1(t) + c_2 \underline{X}^2(t)$
every linear combination of them is a solution

if we have n solutions we can find all

$\{X^i(t)\}$ $i = 1, \dots, n$

If we have n solutions we can find one

$\{\underline{X}^j(t)\}$ $j = 1, \dots, n$

$\underset{n\times n}{M}(t) = \left(\underline{X}_1(t) \dots \underline{X}_n(t)\right)$

$\boxed{\dot{M}(t) = AM(t)}$, because

$\dot{M}(t) = \left(\dot{\underline{X}}^1(t) \dots \dot{\underline{X}}^n(t)\right) = \left(A\underline{X}^1(t), \dots, A\underline{X}^n(t)\right) = A\left(\underline{X}^1(t), \dots, \underline{X}^n(t)\right) = AM(t)$

solving it gives a matrix solution

having a matrix is enough to solve any initial condition

example:
if we have M, $\underline{c}$ constant vector
$M(t)\cdot\underline{c} = c_1\underline{M}^1(t) + \dots + c_n\underline{M}^n(t)$, since the columns solve the system this is a linear combination of them, which is also a solution

Lets find $\underline{X}^0$
$\underline{X}(t)$: $\underline{X}(0) = \underline{x}^0$ we'll look for $\underline{c}$

$M(0)\cdot\underline{c} = \underline{x}^0 \Rightarrow \underline{c} = M^{-1}(0)\underline{x}^0$, M^{-1} needs to exists, $\det M(0) \neq 0$

So for any $\underline{X}(t)$:
$\underline{X}(t) = M(t)M^{-1}(0)\underline{X}^0$

$\underline{X}^j(t)$ – fundamental set of solutions
$M(t)$ – fundamental matrix solution

$\det M(0) \neq 0 \Rightarrow \det M(t) \neq 0$, inverse time or any future solution can be initial

1 dimention: $\dot{X}(t) = aX(t) \Rightarrow X(t) = X_0e^{at}$

$(e^{at})' = ae^{at}$ explaining why this derivative law is true

$e^{at} = \sum_{k=0}^{\infty} \frac{(at)^k}{k!} = 1 + \dots$

$(e^{at})' = \sum_{k=1}^{\infty} \frac{a^k}{k!} \cdot kt^{k-1} = \sum_{k=1}^{\infty} \frac{a^k}{(k-1)!} t^{k-1} = a\sum_{k=1}^{\infty} \frac{(at)^{k-1}}{(k-1)!} \overset{m=k-1}{=} \sum_{m=0}^{\infty} \frac{(at)^m}{m!} = ae^{at}$

general case: $\dot{\underline{X}} = A\underline{X} \Rightarrow \underline{X}(t) = \underbrace{e^{At}}_{2}\cdot\underline{X}_0$

$e^A \overset{\text{def}}{=} \sum_{k=0}^{\infty} \frac{A^k}{k!}$, this will act the same

we had
$\underline{X}(t) = M(t)M^{-1}(0)\underline{X}^0$

we had

$\underline{X}(t) = M(t)\, M^{-1}(0)\, \underline{X}^0$

now we have also

$\underline{X}(t) = e^{At} \cdot \underline{X}^0$



this relies on the uniqueness of the solution, 2 different solutions must emerge from the same point

so $e^{At} = M(t) M^{-1}(0) = \tilde{M}(t)$

Examples

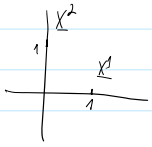
1) simplest case, diagonal matrix

$\begin{cases} \dot{x} = ax \\ \dot{y} = by \end{cases}$ $\quad A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, $e^{At} = ?$

just need to solve for 2 initial conditions

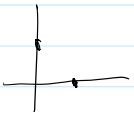
$\underline{X}^1 = \begin{pmatrix} e^{at} \\ 0 \end{pmatrix}$

$\underline{X}^2 = \begin{pmatrix} 0 \\ e^{bt} \end{pmatrix}$ $\quad M(t) = \begin{pmatrix} e^{at} & 0 \\ 0 & e^{bt} \end{pmatrix} = \tilde{M}(t) = e^{At}$



2) harmonic oscilator

$A = \begin{pmatrix} 0 & -w \\ w & 0 \end{pmatrix}$ $\quad \begin{cases} \dot{x} = -wy \\ \dot{y} = wx \end{cases}$



$\underline{X}^1(t) = \begin{pmatrix} \cos(wt) \\ \sin(wt) \end{pmatrix}$ $\quad \underline{X}^2(t) = \begin{pmatrix} -\sin(wt) \\ \cos(wt) \end{pmatrix} \Rightarrow M = \begin{pmatrix} \cos wt & -\sin wt \\ \sin wt & \cos wt \end{pmatrix}$

$e^{\begin{pmatrix} 0 & -w \\ w & 0 \end{pmatrix} \cdot t} = \begin{pmatrix} \cos wt & -\sin wt \\ \sin wt & \cos wt \end{pmatrix}$

3) $C = \underbrace{\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}}_{A} + \underbrace{\begin{pmatrix} 0 & -w \\ w & 0 \end{pmatrix}}_{B}$ $\quad e^{Ct} = ?$

(for numbers $e^{a+b} = e^a \cdot e^b$ proving this uses $a \cdot b = b \cdot a$, not true for matrix)

general case algorithm

$\dot{\underline{X}} = A\underline{X} \Rightarrow X(t) = e^{At} \underline{X}0$

lets look for a solution of shape (V is constant)

$X(t) = e^{\lambda t} \cdot \underline{V}$

$\dot{X}(t) = \lambda e^{\lambda t} \cdot \underline{V}$

$\lambda e^{\lambda t} \cdot \underline{V} = A\underline{X} = A e^{\lambda t} \cdot \underline{V}$

$\lambda \underline{V} = A\underline{V}$

$(A-\lambda I)\underline{V} = 0$ for any V that is non 0, $\det(A-\lambda I) = 0$

$\varphi(\lambda) = \det(A-\lambda I) = 0$ - charactaristic polinomial $\Rightarrow \lambda$

finding λ, Plug it in to $(A-\lambda I)\underline{V} = 0 \Rightarrow \underline{V}$

Plugging λ and V to $X(t) = e^{\lambda t}\cdot \underline{V}$, we get a solution

ther ah be 2 complications:

1) the polinomial might have complex roots

2) might have degenerate roots (e.g. only 2 solutions to a power 3 polinomial)

how to deal with them?

practical example

$\dot{\underline{X}} = \underbrace{\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}}_{A}\underline{X}$

$\varphi(\lambda) = \det\begin{pmatrix} 1-\lambda & 3 \\ 3 & 1-\lambda \end{pmatrix} = (\lambda-1)^2 - 9 = \lambda^2 - 2\lambda - 8$

$\lambda = -2, \quad \lambda = 4$

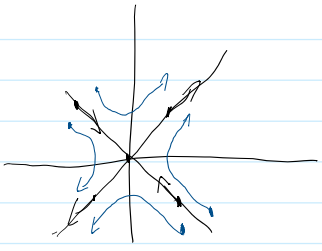
Plug into $(A-\lambda I)\cdot \underline{V} = 0$ $\quad \underline{V} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$

$\left(\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} - \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}\right) V = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}\underline{V} = \begin{pmatrix} 3V_1 + 3V_2 \\ 3V_1 + 3V_2 \end{pmatrix} = 0 \Rightarrow V_1 = -V_2 \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\left(\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}\right) V = \begin{pmatrix} -3 & 3 \\ 3 & -3 \end{pmatrix} V = \begin{pmatrix} -3V_1 + 3V_2 \\ 3V_1 - 3V_2 \end{pmatrix} = 0 \Rightarrow V_1 = V_2 \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Plug these in to $X(t) = e^{\lambda t}\cdot \underline{V}$, to get 2 solutions

$\underline{X}^1 = e^{-2t}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \qquad \underline{X}^2 = e^{4t}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$



complications:

1) complex solution. 2

reminder

$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$

$\det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - bdi - afh$

reminder

$ax^2 + bx + c = 0$

$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

1) complex solution z

$\dot{\underline{z}} = A\underline{z}$, $z(t) = u(t) + iw(t)$ this still fine, since we can look at u and w

every complex solution will have its twin solution $\bar{\lambda} = \alpha - i\beta$ — $a\bar{\lambda}^2 + b\bar{\lambda} + c = 0$

so $\bar{z}(t) = u(t) - iw(t)$ is also a solution

$\lambda = \alpha + i\beta$ — $a\lambda^2 + b\lambda + c = 0$

in complex, these are 2 solutions, but in real, these are the same

example

$\dot{X} = \begin{pmatrix} 0 & 4 \\ -1 & 0 \end{pmatrix} X$ $\quad p(\lambda) = \lambda^2 + 4 = 0 \Rightarrow \lambda = \pm 2i$, $\lambda = \alpha + i\beta$, $\boxed{\alpha = 0 \ , \ \beta = 2}$

$(A - \lambda I)V = \begin{pmatrix} -2i & 4 \\ -1 & -2i \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = 0$ $\quad \begin{cases} -2iV_1 + 4V_2 = 0 \\ -V_1 - 2iV_2 = 0 \end{cases} \Rightarrow V_1 = \frac{2}{i} V_2$

$\underline{V} = \begin{pmatrix} 2 \\ i \end{pmatrix} = \underline{u} + i\underline{w}$

$\boxed{\underline{u} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \underline{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}}$

$(\lambda = \alpha + i\beta)$

$e^{\lambda t}(\underline{u} + i\underline{w}) = e^{\alpha t}\left[(\underline{u}\cos\beta t - \underline{w}\sin\beta t) + i(\cos\beta t \cdot \underline{u})\right]$

$\boxed{e^{i\theta} = \cos(\theta) + i\sin(\theta)}$

$e^{\lambda t}(\underline{u} + i\underline{w}) = e^{\alpha t + i\beta t}(u + iw) = e^{\alpha t} \cdot e^{i\beta t}(u + iw) = e^{\alpha t} \cdot (\cos\beta t + i\sin\beta t) \cdot (u + iw) =$

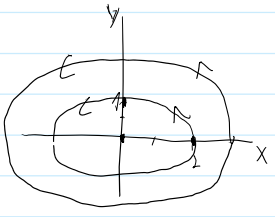
$= e^{\alpha t}\left((u\cos\beta t - w\sin\beta t) + i(u\sin\beta t + w\cos\beta t)\right)$

we take seperately the real and complex parts

in our case

real part $\underline{X}^1(t) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}\cos 2t - \begin{pmatrix} 0 \\ 1 \end{pmatrix}\sin 2t$

imaginary $\underline{X}^2(t) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}\sin 2t - \begin{pmatrix} 0 \\ 1 \end{pmatrix}\cos 2t$



both are real solutions

example

$\dot{X} = \begin{pmatrix} -4 & 5 \\ -5 & 2 \end{pmatrix} X$

$\left|\begin{pmatrix} -4-\lambda & 5 \\ -5 & 2-\lambda \end{pmatrix}\right| = \lambda^2 + 2\lambda + 17 = 0$, $\lambda = \underset{\alpha}{-1} \pm \underset{\beta}{4i}$

$\underline{V} = \begin{pmatrix} 3 - 4i \\ 5 \end{pmatrix} = \underset{u}{\begin{pmatrix} 3 \\ 5 \end{pmatrix}} + i\underset{w}{\begin{pmatrix} -4 \\ 0 \end{pmatrix}}$

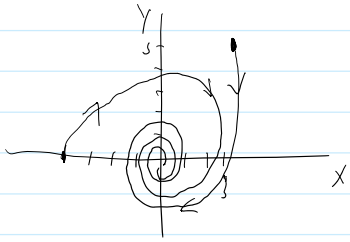
Y_1

$\underline{V} = \begin{pmatrix} 3-4i \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} + i\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

$\underline{u}$ $\underline{w}$

$\underline{X}^1(t) = e^{-t}\left(\cos 4t \cdot \begin{pmatrix} 3 \\ 5 \end{pmatrix} - \sin 4t \begin{pmatrix} -4 \\ 0 \end{pmatrix}\right)$

$\underline{X}^2(t) = e^{-t}\left(\sin 4t \begin{pmatrix} 3 \\ 5 \end{pmatrix} + \cos 4t \begin{pmatrix} -4 \\ 0 \end{pmatrix}\right)$



e^{-t} will win, so it decays towards 0

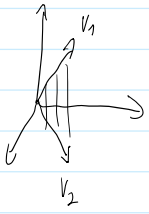
Somehow analyze system at $\begin{pmatrix} 3 \\ 5 \end{pmatrix}, \begin{pmatrix} -4 \\ 0 \end{pmatrix}$ to understand direction

Problem 2, if we dont have enough solutions

$\varphi(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2)^2$

here we have 2 solutions $V^1 e^{\lambda_1 t}$, $V^2 e^{\lambda_2 t}$, but this is 3 dimentional

we can solve things on that plane



there is another vector $\underline{W}$ that satisfies

$(A - \lambda I)W = V_2$

$(A - \lambda I)^2 \cdot W = 0$

$AW = \lambda I W + V$

$\underline{X}(t) = (W + tV_2)e^{\lambda t}$ will be another solution

only V_2, since it is the one with a degenerate λ

$\underline{X}'(t) = W \cdot \lambda e^{\lambda t} + V e^{\lambda t} + tV \cdot \lambda e^{\lambda t} = e^{\lambda t}\left(\lambda W + V + tV\lambda\right) = e^{\lambda t}\left(\lambda(W + tV) + V\right) =$

$= A\underline{X} = A\left((W + tV)e^{\lambda t}\right) = \underbrace{AW}_{\lambda W + V} e^{\lambda t} + \lambda t V e^{\lambda t} = e^{\lambda t}(\lambda W + V + \lambda t V)$